

Exercise Sheet for Week 5 - Type inference and polymorphism

Assigned: Monday 28 October
Suggested Deadline: Monday 4 November

Question 1: Recap: type-checking

Give derivations to show that each of the following is a valid typing judgement (using Curry-style terms).

- (a) $\vdash \lambda x. x (\lambda y. x (\lambda z. y)) : ((\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B}) \rightarrow \mathbb{B}$
- (b) $\vdash \lambda x. (\lambda y. y)x : (\alpha \rightarrow \beta) \rightarrow \alpha \rightarrow \beta$
- (c) $\vdash \lambda f. \text{if } f \text{ true then } f \text{ else } \lambda y. f (f y) : (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B} \rightarrow \mathbb{B}$

Question 2: Type-inference, or finding principal types

For each of the following closed Curry-style λ -terms, find its principal type.

- (a) $\lambda x. x (\lambda y. x (\lambda z. y))$
- (b) $\lambda x. (\lambda y. y)x$
- (c) $\lambda f. \text{if } f \text{ true then } f \text{ else } \lambda y. f (f y)$
- (d) $\lambda x. \lambda f. \text{if } f x \text{ then } f \text{ else } f$

Question 3: Type-checking let-polymorphism

Give derivations to show that each of the following is a valid typing judgement (Curry-style λ -calculus with let-polymorphism).

- (a) $x : \forall \alpha. \alpha \vdash x : \mathbb{N}$
- (b) $x : \forall \alpha, \beta. \alpha \vdash x : \mathbb{N}$
- (c) $f : \forall \alpha. \alpha \rightarrow \alpha \vdash \text{if } f \text{ true then } f \text{ else } \lambda x. x : \mathbb{N} \rightarrow \mathbb{N}$
- (d) $f : \forall \alpha. \alpha \rightarrow \alpha \vdash f f : \beta \rightarrow \beta$
- (e) $e : \forall \alpha. \alpha \rightarrow \alpha \rightarrow \mathbb{B}, x : \beta, y : \beta \vdash \text{if } e x y \text{ then succ zero else zero} : \mathbb{N}$
- (f) $\vdash \text{let } f = \lambda x. \lambda y. x \text{ in if } f \text{ true false then } f \text{ zero true else } f \text{ zero } f : \mathbb{N}$

Question 4: type-inference for let-polymorphism

For each of the following terms (Curry-style λ -calculus with let-polymorphism) find a type, type environment, and typing derivation, or else show that none exists. Try to find the most general type in each case.

- (a) $x y$
- (b) `if iszero  $x$  then  $f x$  else  $f$  true`
- (c) $\lambda f. \text{succ}(f x)$
- (d) `let  $f = \lambda x. x x$  in  $f (\lambda y. y)$  true`
- (e) `let  $f = \lambda x. x$  in  $f (\lambda y. y)$  true`

Question 5: Type-checking System F terms

Give derivations to show that each of the following is a valid typing judgement (Church-style System F terms).

- (a) $x : \mathbb{N} \vdash (\lambda f : \forall \alpha. \alpha \rightarrow \alpha. f \{\mathbb{N}\} x) (\lambda \alpha. \lambda y : \alpha. y) : \mathbb{N}$
- (b) $\vdash \lambda \alpha. \lambda x : \alpha. \lambda y : \alpha. x : \forall \alpha. \alpha \rightarrow \alpha \rightarrow \alpha$
- (c) $x : \alpha \vdash \lambda \gamma. \lambda f : \alpha \rightarrow \gamma. \lambda g : \beta \rightarrow \gamma. f x : \forall \gamma. (\alpha \rightarrow \gamma) \rightarrow (\beta \rightarrow \gamma) \rightarrow \gamma$
- (d) $\vdash \lambda f : \forall \alpha. \alpha. f \{\beta\} : (\forall \alpha. \alpha) \rightarrow \beta$

Question 6: Small-step CBV semantics of System F terms

Reduce each of the following (Church-style, System F) terms to a CBV normal form.

- (a) $(\lambda f : \forall \alpha. \alpha \rightarrow \alpha. f \{\beta\}) (\lambda \alpha. \lambda x : \alpha. x)$
- (b) $(\lambda g : \forall \gamma. (\mathbb{N} \rightarrow \mathbb{B} \rightarrow \gamma) \rightarrow \gamma. g \{\mathbb{N}\} \lambda x : \mathbb{N}. \lambda y : \mathbb{B}. x) (\lambda \gamma. \lambda f : (\mathbb{N} \rightarrow \mathbb{B} \rightarrow \gamma) \rightarrow \gamma. f \text{ zero true})$